Automatically Making Origami Diagrams

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Abstract. A traditional way to explain the folding process of origami works is usually using a sequence of diagrams (called origami illustrations). Since making origami illustrations is a heavy burden for origami designers, new original works are difficult to be published in time. To solve this problem, we propose a unique framework for automatically making origami illustrations based on 2-D crease patterns, and moreover give the key techniques to fully support proposed framework.

Keywords: Origami diagram making, graph recognition, transform form 2-D crease pattern to 3-D origami model.

1 Introduction

Origami is popular art of folding paper. In its long history, not only a large number of traditional works have been accumulated, but many original works are designed and published into origami drill books every year. For most of origami lovers, the origami drill books [1], in which a folding process for a piece of work is explained by a sequence of diagrams (called origami illustrations), are probably the most convenient means for learning and enjoying origami making. Recent years, accompanied with the Internet spreads, considerable sum of original origami works come to announce via web pages, but there is no change in the point to explain folding process by a series of origami illustrations (e.g. [2], [3]).

However, making origami illustrations requires a large amount of labor. Since it is a heavy burden for designers, new origami works are usually difficult to be published in time. To solve this problem, this paper proposes an interactive supporting framework that makes it possible to realize automatically making origami illustrations. Although many researches on origami simulation, design, etc. have been conducted recently, such a framework has never been reported.

2 Basic Idea

A straightforward approach to automatically making origami illustrations is probably to simulate the folding process of origami works. That means the system needs to always keep the 3-D models of origami works, and moreover to update the models every time folding operations are applied. Obviously, that is a too tough task to be achieved.

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To avoid keeping and manipulating the 3-D models all the time, our approach is originally based on 2-D crease patterns. The basic idea is to utilize the 2-D crease patterns as much as possible, and transform the crease patterns into 3-D models only in unavoidable instances. Therefore, as shown in Fig.1, the concept of our system can be explained by a circular model where information is enhanced from 2-D (crease patterns) to 3-D (3-D origami model); than the 3-D information is further reduced into 2.5-D (origami illustrations) and finally back to 2-D (crease patterns). A big advantage of our approach is that when a folding operation is applied, instead of updating the 3-D origami model, we need just to add some creases on the 2-D crease pattern so that it can correspond to origami's updated state. Obviously, the latter is much easier than the former.

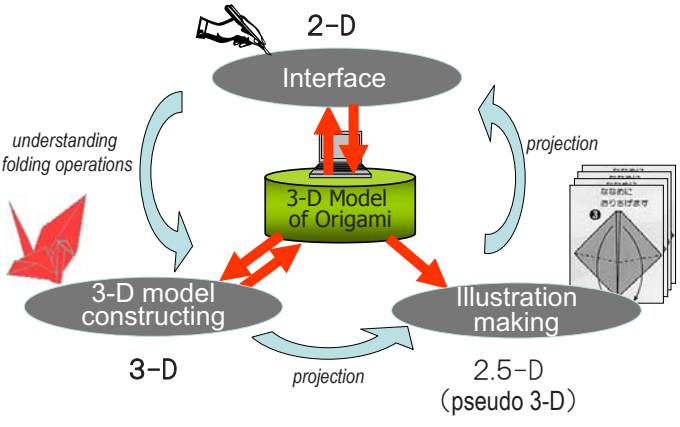
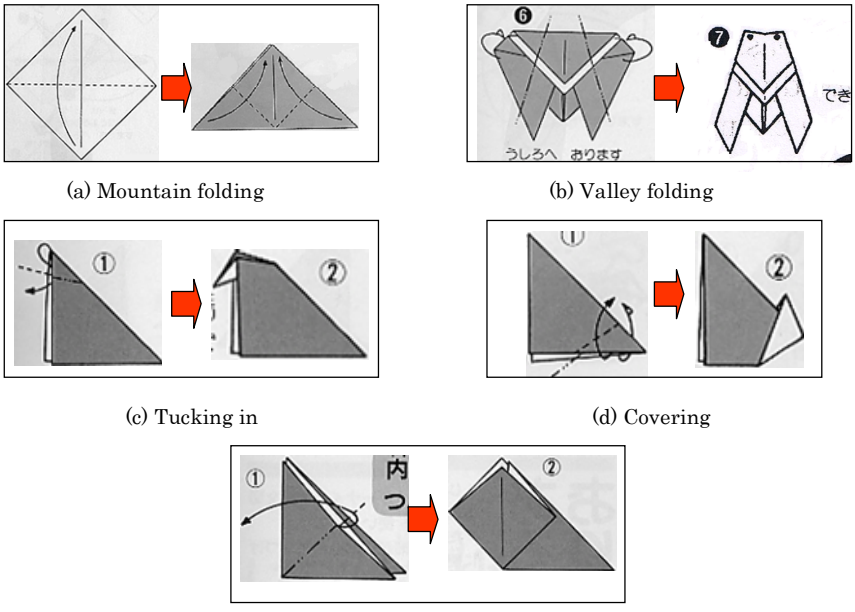


Fig. 1. Circular model for information processing and transformation

Some key techniques to support the basic idea have been developed. First, we have constructed an interface which allows the designers to communicate with the system by freehand drawing. That is, origami designers can input origami illustrations by pen (initially) or draw symbols on existing illustrations to tell the system which operation will be done next. The interface can understand the users' intention. Second, we have also developed algorithms to reflect the users' intention into a foldable crease pattern (sometimes, multiple crease patterns). Third, we have established a method to transform creases patterns into 3-D origami models using computer graphic techniques. To make origami illustrations, we just need to project the 3-D origami models from an appropriate perspective.

3 Interactive User Interface

We use the notation of origami illustrations proposed by Yoshizawa [4] that has already become a kind of international standard. There are five kinds of oft-used folding operations. They are divided into basic folding ("mountain", "valley" folding) and complicated folding ("tucking in", "covering", and "opening"). Fig. 2 shows



(a) Mountain folding (b) Valley folding

(c) Tucking in (d) Covering

(e) Opening

Fig. 2. Five kinds of frequently used origami operations and their notations in origami dill books

some illustrations of these operations. From Fig.2 (a)-(e), we can find that recognition of these operations mainly depends on if we can recognize some handing-written straight lines and curves. Concretely, it is necessary to recognize two types of straight lines (solid/broken), arrows (solid/hollow), arcs (C/S-type) and also necessary to make use of the stroke orders of theses symbols.

Our methods to recognize these symbols are very simple. Given a hand-written curve, we first draw an axis that links the beginning and ending of the curve with the direction from beginning toward the ending. Then, we integrate the curve along the axis. Let the parts of positive and negative integration be d^+ and d^-. For example, the C/S-type arcs and straight lines can be distinguished in the way described as follows.

1. The curve is recognized as a C-type arc (Fig.3 (a)), which is used for describing operations excluding "tucking in", if they meet: $d^+ > TH_{arc_right}$
2. The curve is recognized as straight line (Fig.3(b)), if they meet: $\left|d^+\right| + \left|d^-\right| < \varepsilon$.
3. The curve is recognized as S-type arc (Fig.3 (c)), which is used for describing "tucking in", if they meet: $d^+ + d^- < \varepsilon$.

Here, TH_{arc_right} and ε are a large and small thresholds. The folding operations are finally determined based on individual symbols and their spatiotemporal relations.

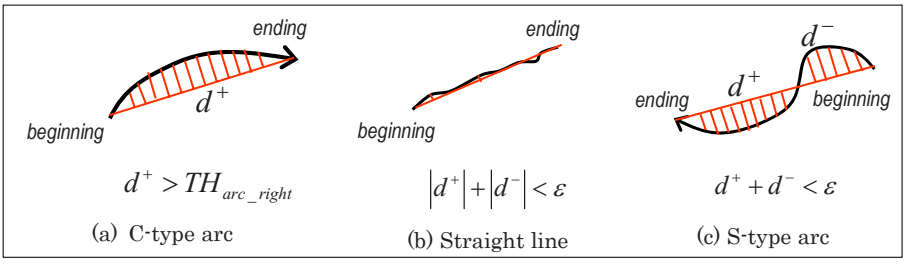


Fig. 3. Recognition of symbols that express folding operations

4 Generation of Foldable Crease Patterns

According to information about the type and location of the folding operation obtained via the user interface, the crease pattern is updated to meet the uses' intention. We summarize our method to generate foldable crease patterns as follows.

The creases caused by a basic folding can be generated based on the following features:

(1) **Connectivity:** the terminal points of a crease locate either on an outside-edge of the square (virtual origami sheet) or share an inner point with another crease caused by the same fold.

(2) **Symmetry:** when a terminal point of a crease (A) reaches to an existing one (B), a new crease (C) extends from the connective point in the direction that makes creases A and C symmetrical against crease B. Creases A and C have opposite attributes (mountain/valley).

Fig.4 shows an example of generating creases made by a basic folding applied to a simple origami model. As to origami models in complicated states, the creases can be generated by repeatedly making use of feature (2) until a terminal point reaches to an outside-edge of the square or no more new creases can be generated (stop conditions).

We generate creases caused by a complicated folding using their own features and foldablity theorem called Local Flatness Conditions [5], [6] to limit the number of the folds we need to take into account. "Tucking-in" and "covering" have some identical/different features with each other that are described as follows.

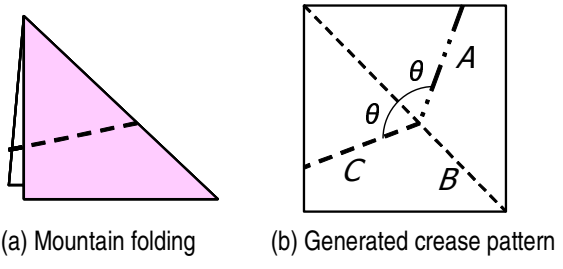


Fig. 4. Calculation of crease patterns for basic folding

(1) **Identical Features:**

- Generated creases (A and B) are symmetrical to an existing crease (called symmetrical axis).
- The symmetrical axis is separated into two segments, and the attribute of one piece (C) is reversed.

(2) **Different Features:**

- Generated creases have the same (opposite) attribute with (from) the symmetrical axis for tucking-in (covering).
- The creases generated by tucking-in (covering) form an acute (obtuse) angle with the symmetrical axis.

The creases caused by "tucking-in" and "covering" are simply generated based on above features in two steps. First, the new creases are made on the position given by a fold-line, using the method to make creases for basic folding. Second, the attribute of a part of the symmetrical axis is reversed. The way to make creases for "opening" can be referred in [7]. Some examples can be seen in Figs.5.

5 Construction of 3-D Origami Model

We construct the 3-D origami model based a foldable crease pattern using Affine transformation. Affine transformation relies on the neighboring relationships among faces (polygons in crease patterns) and the information attached to creases. A face-crease graph (CP-graph), where the nodes and links indicate the faces and creases, is thus introduced to easily grasp these face relationships and crease information. In Fig.6, we show a crease pattern (left), the face-crease graph (middle) and expression of a crease (right).

(1) Formalization

From the viewpoint of easily constructing 3-D origami models, a crease is specified by both of its position(coordinates) and the angle $\theta\ (-\pi \le \theta \le \pi)$ between two faces that share the crease. The crease $C_{i,j}$ is specified by the unit vector $\hat{c}_{i,j}$ (clockwise to F_i), and the vector $t_{i,j}$ that links the origin and the beginning of $C_{i,j}$ with the direction from the former to the latter.

(2) Affine Transformation

An arbitrary face F_0 in the crease pattern is fixed on x-y plane. We calculate the transformation matrix for other faces (notated by F_p) with two steps. First, a path, the shortest one is desirable, is found out from F_0 to F_p . Let the path be

$$F_0, \cdots, F_q, \cdots, F_p . \tag{1}$$

So, the creases on this path can be indicated by

$$C_{0,1}, \cdots, C_{q-1,q}, \cdots, C_{p-1,p} . \tag{2}$$

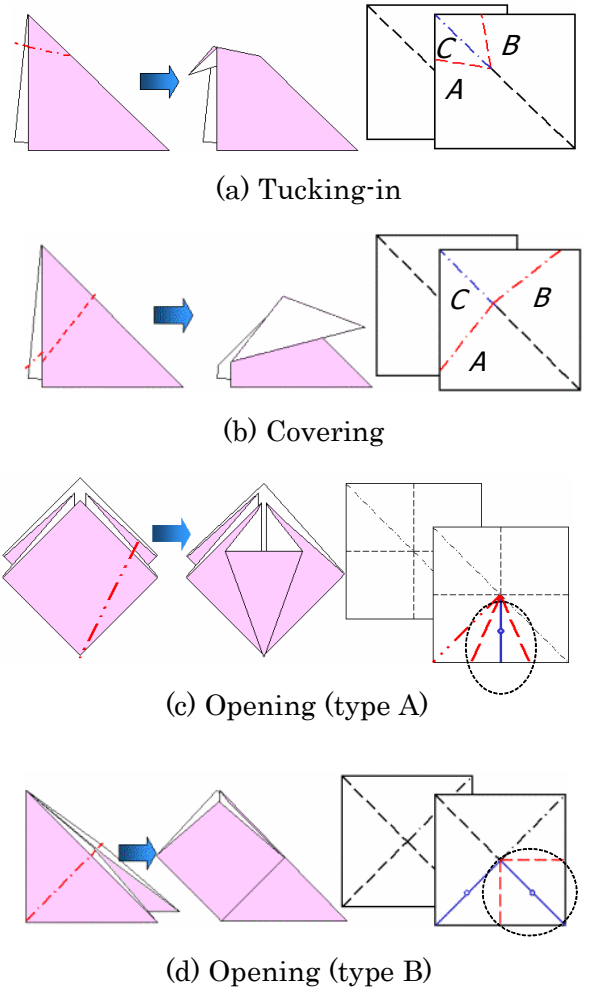


Fig. 5. Calculation of crease patterns for complicated folding

Then, the transformation matrix based on the sequence of creases shown in above is calculated. Let rotation matrix in association with angle θ of $\hat{c}_{q-1,q}$ be R_q, and translation matrix in association with vector $t_{q-1,q}$ be T_q. The Affine transformation matrix for crease $C_{q-1,q}$ can be written as

$$X_q = T_q R_q T_q^{-1}. \tag{3}$$

So, the 3-D Affine transformation matrix for face F_p along the path we have found out can be calculated by

$$Z_q(x, y, 0, 1) = X_1 \ldots X_q \ldots X_p (x, y, 0, 1)^T \text{ for } (x, y) \in F_p. \tag{4}$$

By such a transformation for each face in the crease pattern, it is possible to constitute the 3-D origami model. We show an example in Fig.7.

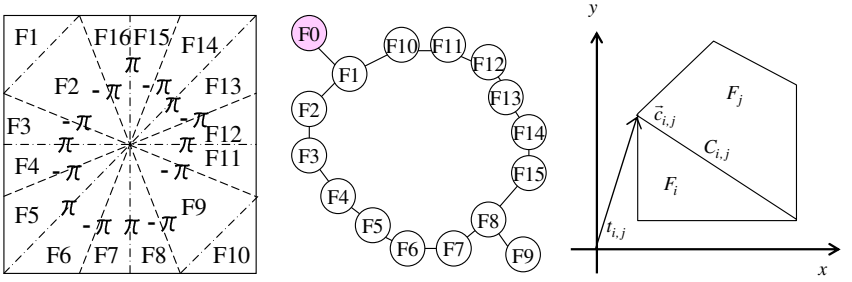
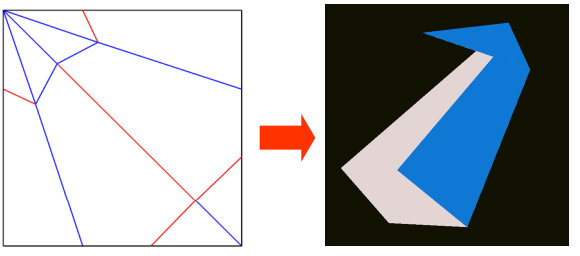


Fig. 6. An example of crease patterns (left), FC-graph (middle) and formalization of a crease



(a) Crease pattern (b) 3-D origami model

Fig. 7. An example of constructing the 3-D origami model based on a 2-D crease pattern

6 Conclusions

In this paper, we have proposed a framework for automatically making origami illustrations with the viewpoint to accelerate publication of origami works. We also described the key techniques that fully support the proposed framework. The experiments for testing individual units of the system have shown very good results. As the future work, the integration of different units has to be completed.

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